

between conductors, that touch sensors utilise to detect a touch.

How it works

Human beings, and in fact the majority of carbon based life, are composed of a variety of chemical compounds mixed with a large quantity of water. Water is a fluid composed of electrically-polarised molecules, which means, an electric field can polarise molecules in water very easily. As a result, water based and carbon based life have a very high relative permittivity ($\epsilon_r > 60$), so there is a very pronounced effect on electric fields. That is what allows the usage of capacitive-touch interfaces (Fig. 2).

Typically, human beings show up as an increase in the capacitance of a sensor. So, all we need for a capacitive-touch or capacitive-proximity interface is a method for measuring the capacitance of the sensor to a sufficient resolution and a conductive sensor pad. (Note: The other conductor of the sensor is typically the electrical ground of the circuit, and the actual conductor is the ground of the circuit. For line-operated systems, this ground is typically earth-ground. So, line-operated systems actually enjoy greater touch sensitivity because we are in close proximity to a lot of earth.)

The challenge

So, all we need is a means to measure capacitance, right? Well, in a perfect world, yes! Unfortunately, we live in an imperfect and somewhat noisy world, so we actually have to add a few qualifiers to that statement. What we actually need is a method for measuring capacitance that is both low-impedance and has low-susceptibility to noise. The low-impedance part prevents external electric fields (conducted noise) from affecting the capacitance measurement, and the low-susceptibility to noise prevents external RFI (radiated noise) from affecting the capacitance measurement.

Conducted noise. If the capacitance-measurement system has high levels of conducted, common-mode

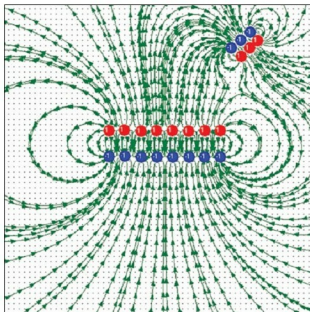


Fig. 2: An example plot of the electric field lines disturbed by high ϵ_r material

noise on its power supply, it will look like noise is being injected into the touch-sensor. Remember that the circuit cannot tell the difference between it moving up and down electrically, and the sensor being drug up and down electrically. So, conducted noise looks to the circuit like noise on the sensor. By using a low-impedance measurement system, we reduce the effects of conducted noise by dragging water molecules of the user up and down in time with circuit-ground, and limit the effect of the ambient-ground pulling on our sensor through the user.

Removing external noise. To subtract out the external noise, we typically use a differential-measurement method, if possible. While it would be great if we could tap into the user's ground for this function—it is generally problematic to connect to the user's ground. So, instead, we try to do two measurements, one with a positive charge on the sensor and one with a negative charge. When we subtract the two, we get an approximation of a differential measurement that is good for most low-frequency noise.

Limiting radiated noise. To limit the effects of radiated noise, we do two things—we limit the amount of time that the sensor is connected to the conversion circuitry, providing

a pathway for the noise, and we dither the timing of the sample to prevent a beat frequency between the sample rate and the radiated noise. Actual mechanics of the conversion process also determine the susceptibility of the system to radiated noise, so some conversion methods will be better suited to the rejection of radiated noise.

Other challenges

All these techniques help reduce the amount of noise coupled into capacitance conversion. However, no matter how carefully we

make the conversion, some noise will sneak through in the sample. In addition, the amount of shift we see due to a touch, and particularly proximity detection, is quite small.

To handle the noise that does come through and to improve sensitivity, we take multiple samples and average the results together. This increases the amount of shift we see due to a touch, helps average out the noise and actually limits the rate of change in the measured value. After all, the speed of a user's touch is significantly lower than the typical frequency of noise in the system, so we can live with a system that has a slower response time if it helps to cancel out the noise.

Another useful function is to incorporate a slew-rate limiter on the data. Basically, this function looks at each new sample; if the sample is above average, the average is increased by 1-5, and if it is less, the average is decreased by a similar amount. This prevents large noise spikes from dragging our average up and down, while still passing slower changes in the samples.

Together, these functions allow a capacitive-touch system to operate, even in noisy environments. It turns out, that is exactly the environment in which lighting systems have to be able to survive. After all, lighting systems typically share power with some prolific noise